Section	offset	name	nBytes	Comments
Game Object	0	x coord world	4	
	4	y coord world	4	
	8	z coord world	4	
	12	Roll Angle	2	32768=pi radians
	14	Pitch Angle	2	
	16	Yaw Angle	2	
	18	Distance near/far code	1	3=far far away, 2= med far, 1=med near, 0=near (used for Level-of-Detail)
	19	Class Code	1	objhq.class=3, obj5da(plane).class=1, objshard.class = 2, objrv(rescue vehicle).class = 1, bomb.class = 0
	20	Distance from camera (squared)	4	Probably used to sort objects so they are drawn in the correct order
	24	object type	2	0=invisible, 0x8=non destructable, 0x800=on hold, 0x104=crashed plane, 0x10=timed object
	26	rotflag	1	0=no rotation, 2=rotate until med far, 3= rotate until far far away (see distance code)
	27	hitpoints	1	
	28	Bkpt for d^2 near	4	Compare dist to this number to be considered 'near' for level of detail (LOD)
	32	Bkpt for mednear	4	
	36	Bkpt for med far	4	
	40	Model to use for near	2	16bit memory pointer to model
	42	model for med near	2	16bit ptr
	44	model for med far and far	2	16bit ptr
	46	x velocity	2	signed 16bit
	48	y vel	2	
	50	z vel	2	
	52	timer value	2	When gets to zero, timed objs are deleted
	54	roll angular vel	2	
	56	pitch angular vel	2	
	58	yaw angular vel	2	
	60	dispersion	2	Has to do with randomness of path thru space
	62	damping	2	How air resistance slows this down
	64	Gravity	2	gravity effect on this object
	66	x-size	2	collision rectangle x size
	68	y-size	2	collision rectangle y size
	70	z-size	2	collision rect z size
	72	function ptr for tmr callback	2	Ptr into objtimtable in defobjs.asm. This function is called every frame?

Section	offset	name	nBytes	Comments
3D Model Object	0	x coord	2	local x coord
	2	y coord	2	
	4	z coord	2	
	6	Num primitive models	2	Models are made of a number of primitives. Each has their own structure. The array of primitives is tacked onto the end of this structure following this item
	8	Primitive class	2	See cmpjtbl in f3dvec.asm. 0=no draw, 2=wireframe, 4=point cloud, 6=sphere, 8=polygon
	10	Ptr to primitive model data	2	Memory ptr to data for primitive. Different for each type of primitive
	12	(next primitive class)	2	
	14	next ptr to prim data	2	etc etc etc.
	16	.... repeats		
Null Primitive Data	--			No data for Null primitive
Frame Primitive Data	0	Number of wires	2	A frame primitive is made up of a number of wires.
	2	ptr to first wire	2	16 bit ptr to a Wire primitive structure - See wire subprimitive structure below
	4	ptr to 2nd wire	2	
	6	ptr to 3rd wire ...	2	
	8	...		

Section	offset	name	nBytes	Comments
Point Cloud Primitive Data	0	Number of points	2	Number of points in the cloud
	2	Color of points	2	If <256, color is constant as specied. Otherwise, this is a 16bit ptr to a color code
	4	X coord local (pt0)	2	
	6	Y coord Local (pt0)	2	
	8	Z coord Local (pt0)	2	
	10	X coord Local (pt1)	2	
	12	(... and so on)	2	

Sphere Primitive Data	0	Number of Spheres	2	A sphere primitive is made up of a number of spheres, each can have a different size and color
	2	X coord Local (sph0)	2	
	4	Y coord local (sph0)	2	
	6	Z coord local (sph0)	2	
	8	Sphere Radius (sph0)	2	Actually diameter is used!
	10	Sphere color (sph0)	2	direct color or 16bit ptr to color
	12	X coord local (sph1)	2	
	14	(...)		

Section	offset	name	nBytes	Comments
Poly Primitive Data	0	Number of edges in poly	2	Number of sides must be 3 or greater
	2	Color of polygon	2	direct color or 16bit ptr to color
	4	X coord local (pt0)	2	
	6	Y coord local (pt0)	2	
	8	Z coord local (pt0)	2	
	10	X coord local (pt1)		
	12	(...)		There is one point for each vertex in poly.

Wire Sub-primitive	0	Number of segments in wire	2	There are n+1 points in a wire that has n segments
	2	Color of wire	2	direct color or ptr to color
	4	X coord local (seg0)	2	
	6	Y coord local (seg0)	2	
	8	Z coord local (seg0)	2	
	10	X coord local (seg1)	2	
	12	(...)	2	

notes: